MATHEMATICS

University *of* NORTH ALABAMA

Fall 2025 Math Problem of the Week: Problem #9

Due noon Monday, October 27
Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, October 27 or, if you cannot find me, you may leave it with Ms. Phifer in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #9: Let

$$\begin{array}{cccc} a_{1,1} & a_{1,2} & a_{1,3} & \cdots \\ a_{2,1} & a_{2,2} & a_{2,3} & \cdots \\ a_{3,1} & a_{3,2} & a_{3,3} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{array}$$

be a doubly indexed sequence of positive integers such that each positive integer appears exactly eight times in the sequence. Prove that $a_{m,n} > mn$ for some pair of positive integers (m, n).